



**COMSATS University Islamabad,
Sahiwal Campus, Pakistan**

Hope Givers

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By

Ayesha Ishfaq

M Jawad Haider

Bilal Hassan

CIIT/SP22-BCS-031/SWL

CIIT/SP22-BCS-140/SWL

CIIT/SP22-BCS-155/SWL

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Ayesha Ishfaq

M Jawad Haider

Bilal Hassan

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It is to certify that the final year project of BS(CS) “**Hope Givers**” was developed by **Ayesha Ishfaq (CIIT/SP22-BCS-031/SWL)**, **M Jawad Haider (CIIT/SP22-BCS-140/SWL)** and **Bilal Hassan (CIIT/SP22-BCS-155/SWL)** under the supervision of “**Dr. Yawar Abbas**” and co supervisor “**Mr. Arslan Sarwar**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Science.

Supervisor

Dr. Yawar Abbas

Lecturer, Department of Computer Science
COMSATS University Islamabad, Sahiwal Campus

External Examiner

Head of Department

Dr. Zafar Iqbal Roy

Assistant Professor

Department of Computer Science
COMSATS University Islamabad, Sahiwal Campus

Executive Summary

Hope Givers is a web-based donation and campaign management system designed to connect donors with verified NGOs in a transparent and secure way. The main purpose of the system is to solve the biggest problem in online donations: lack of trust and clarity about how funds are used.

The platform allows NGOs to register, submit verification documents, and create donation campaigns. Donors can browse campaigns, view details, donate securely, and track the progress of each cause. To improve trust and transparency, Hope Givers includes an AI-powered chatbot that provides instant answers about campaign purpose, fund usage, past updates, and future plans using verified data.

The system follows a three-tier architecture presentation layer, business logic layer, and data layer to ensure security, scalability, and easy maintenance. Using Agile methodology, the development is completed through iterative sprints, enabling continuous improvements, feature updates, and stakeholder feedback.

Overall, Hope Givers offers a reliable, user-friendly, and transparent donation environment that strengthens trust between donors and NGOs while ensuring secure and well-managed contribution processes.

Acknowledgment

All praise is to almighty Allah who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

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Ayesha Ishfaq

M Jawad Haider

Bilal Hassan

Abbreviations

CUI	COMSATS University Islamabad
SRS	Software Requirements Specification
SDD	Software Design Description
UC	Use Case
HCI	Human-Computer Interaction
DBMS	Database Management System
UT	Unit Testing
FT	Functional Testing
IT	Integration Testing
NoSQL	NoSQL (Flexible Database Format)
IDE	Integrated Development Environment
CRUD	Create, Read, Update, Delete

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Chapter 1

Introduction

1 Introduction

Hope Giver is a web-based Donation Management System designed to provide a transparent, secure, and reliable connection between donors and verified NGOs. Many donors hesitate to contribute to online campaigns because they are unsure where their money goes or whether the campaign is authentic. Hope Giver solves this issue by offering NGO verification, clear campaign details, donation tracking, and an AI-powered chatbot that explains how funds are used.

The system provides a structured platform where NGOs can create campaigns, upload documents, and share updates. Donors can view goals, see collected amounts, read progress information, and donate safely. The admin oversees the entire system approving NGOs, monitoring campaigns, and ensuring transparency.

Hope Givers is designed using modern web technologies and a three-tier architecture, making the system scalable, secure, and easy to maintain. The use of Agile methodology helps in development through small iterative steps, ensuring continuous improvement and user feedback.

Overall, Hope Givers aims to build trust between NGOs and donors by improving clarity, reducing fraud, and creating a smooth and interactive experience for all users.

1.1 Brief Overview

Hope Givers is a modern, web-based platform designed to make the donation process transparent, trustworthy, and easy for everyone involved. The system allows verified NGOs to create donation campaigns, upload supporting documents, and share progress updates. Donors can browse these campaigns, check the goals and collected amounts, and donate securely.

One of the most important features of Hope Givers is its AI-powered chatbot, which provides instant answers about campaign purpose, fund usage, donation impact, and future plans. This ensures that donors always have clear and verified information before contributing.

The platform follows a three-tier architecture—Presentation Layer (frontend), Business Logic Layer (system operations), and Data Layer (database). This structure makes the system scalable, well-organized, and secure. The development process uses the Agile methodology, allowing the project to be built in small iterations, with continuous feedback and improvements.

In short, Hope Givers provides a reliable environment where NGOs can manage campaigns easily, donors can contribute confidently, and the admin can maintain full oversight of system activities.

1.2 Relevance to Course Modules

The development of Hope Givers is closely connected to multiple Computer Science course modules studied during the degree. Each part of the system applies concepts learned in different subjects, making the project a practical implementation of academic knowledge.

Web Development:

Hope Givers is a web-based system, so frontend and backend concepts from web development courses are heavily applied. This includes HTML/CSS/JavaScript for interface design and PHP/Django/Node (as used in your SDD) for backend logic.

Software Engineering Principles:

The entire project follows Software Engineering principles, including requirement analysis, SDD/SRS documentation, architectural design, and testing. Agile methodology and iterative development directly relate to the course content.

Database Systems:

The system stores user accounts, NGO data, campaigns, donations, and chatbot knowledge using database concepts such as tables, relationships, normalization, CRUD operations, and secure data handling.

Artificial Intelligence:

The AI-powered chatbot uses basic NLP-style classification and knowledge-based response matching. These concepts relate to AI course modules.

UI/UX Design:

A useDesigning a clean, simple, user-friendly interface applies HCI principles like usability, accessibility, clear navigation, and responsive design.

Project Management:

Planning sprints, managing tasks, dividing modules, and tracking progress all follow project management concepts.

Networking

Donor information, payment data, passwords, and NGO documents must be protected. Concepts like hashing, authentication, authorization, and secure transactions relate to security courses.

1.3 Project Background

In recent years, global humanitarian crises and local emergencies have increased the need for transparent and trustworthy donation platforms. Many donors want to help, but they hesitate because most online campaigns do not provide clear details about how the donated money is used. People often worry about fraud, misuse of funds, and fake campaigns.

At the same time, NGOs face difficulties in reaching the right donors and proving that their campaigns are genuine. They struggle to provide regular updates, show transparency, and manage donations manually. This creates a gap between donors who want to help and NGOs who need support.

To solve this problem, Hope Givers was created.

The system provides:

- A secure platform where donors can view real, verified campaigns.
- A document verification process for NGOs to ensure authenticity.
- A clear donation tracking system showing goals and collected amounts.
- A progress update feature so donors can see how funds are used.
- An AI-powered chatbot that explains campaign purpose, fund usage, and updates instantly.

Hope Givers goal is to build trust, remove doubts, and create a transparent bridge between donors and NGOs. By using modern technology and clear verification methods, it ensures that help reaches those who truly need it.

1.4 Literature Review

Table 1: Literature Review

Application Name	Weakness	Proposed Project Solution (Hope Givers)
General Donation Platforms	Lack of deep verification for NGOs Limited clarity on actual fund usage Donors must rely on campaign owners' updates.	NGO verification system Clear display of fund goals & collected amounts Documented campaign updates
NGO Websites	Limited reach Manual donation process No central place for all campaigns Weak transparency reporting	Centralized platform with multiple NGOs Automated donation recording Real-time progress updates All campaigns in one place
Charity Aggregator Websites	Display basic information only No chatbot support or instant answers Limited donor interaction features	AI-powered chatbot for verified info Donor-friendly UI Detailed campaign structure
Social Media Fundraising	High risk of fake or unverified campaigns No tracking or proper records No system-driven transparency	NGO document verification Admin approval of campaigns Donation logs, audit trails, and updates

1.5 Analysis from Literature Review

The literature review highlights several key issues in existing donation and fundraising systems. Most platforms lack strong NGO verification, detailed fund usage transparency, and structured donor communication. Systems like social media fundraising carry high risks of scams and unverified campaigns, while general donation platforms provide limited information beyond basic campaign descriptions.

Hope Givers addresses these gaps by combining verification, transparency, and automation into one platform. The system ensures that NGOs are properly validated through document

submission and admin approval. Donors receive clear details about campaign goals, collected amounts, and progress updates. Unlike existing platforms, Hope Givers includes an AI-powered chatbot that provides instant, accurate, and verified information about campaign usage and donor impact.

Overall, the analysis shows that Hope Givers fills the transparency and trust gap that current systems fail to solve, offering a safer, clearer, and more structured donation experience.

1.6 Methodology and Software Lifecycle

The development of Hope Givers follows a structured and flexible approach to ensure smooth progress, clear planning, and continuous improvement. The chosen methodology is Agile, supported by the Iterative Model, which matches the project's needs for transparency, regular updates, and feature expansion.

1.6.1 Rationale behind Selected Agile Model

Agile is used because it allows the system to be developed in small, manageable parts called *sprints*. After every sprint, feedback can be taken from users, supervisors, or developers to improve the next version.

Agile fits Hope Givers because:

- Requirements can change as new ideas appear.
- Testing happens regularly instead of waiting till the end.
- Features like campaigns, donations, and chatbot can be built step-by-step.
- Easy to add or improve features such as verification, updates, chatbot responses, etc.

1.6.2 Iterative Model

The iterative model supports Agile by dividing the project into repeated cycles. Each cycle improves the system and adds new features.

For Hope Givers, the iterations are:

Iteration 1:

User authentication, basic dashboards, NGO registration.

Iteration 2:

Campaign creation, donation processing, progress updates.

Iteration 3:

AI chatbot integration, knowledge base, campaign explanation.

Iteration 4:

Final optimization, admin logs, security improvements.

This combination ensures the system becomes more complete, secure, and user-friendly after each cycle.

1.7 Summary

This chapter introduced the purpose and need for the Hope Givers donation system. It explained how the platform provides transparency, NGO verification, secure donations, and chatbot-based guidance. The literature review highlighted problems in existing systems, and the analysis showed how Hope Givers solves them. Finally, the Agile and Iterative models were selected to ensure continuous improvement and flexible development.

Chapter 2

Problem Definition

2 Problem Definition

This chapter discusses the precise problem to be solved. It should extend to include the outcome.

2.1 Problem Statement

Many donors want to support important causes, but they hesitate because most online donation systems lack transparency and proper verification. Donors often cannot see how their money is used, which creates trust issues and fear of fraud. NGOs also struggle to reach donors and provide regular updates. There is no centralized platform where verified NGOs, clear campaigns, and transparent donation tracking are combined together.

Hope Givers aims to solve these problems by providing a secure, verified, and transparent donation platform for both donors and NGOs.

2.2 Deliverables and Development Requirements

User Interface: A simple and responsive web UI for donors, NGOs, and admin.

User Registration: Secure login and role-based access for all users.

NGO Verification: NGOs upload documents for admin approval before creating campaigns.

Campaign Module: NGOs can create campaigns, add details, and post updates.

Donation System: Donors can donate securely, and amounts update in real time.

AI Chatbot: Provides quick answers about campaigns, fund usage, and progress. Search & Filters: Donors can search campaigns by category or urgency.

Security: Encrypted data, secure authentication, and privacy protection.

Scalability: System designed to support growth in NGOs, donors, and campaigns.

Testing & Documentation: Proper testing, manuals, and developer documentation included.

2.3 Current System

Currently, there is no centralized and verified platform where donors can safely browse campaigns or track how their donations are used. Most donations are made through social media posts, personal messages, or unverified links, which increases the risk of fraud. NGOs also struggle to manage campaigns, provide updates, and reach potential donors effectively. Due to the lack of transparency, donors often remain unsure about the authenticity of campaigns and the actual usage of funds.

Chapter 3

Requirement Analysis

3 Requirement Analysis

3.1 Use Case Diagram

In this Use case diagram, we show the behavior of our system. This enables you to visualize the different types of roles in a system and how those roles interact with the system.

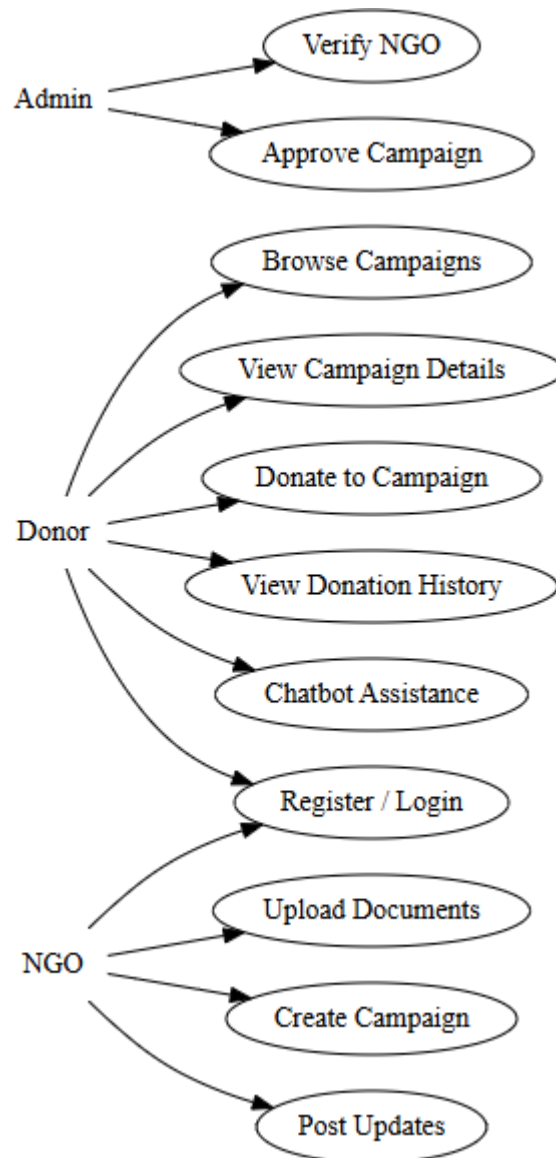


Figure 3.1: Hope Givers Use Case Diagram

3.2 Use Cases Descriptions

The table below indicate comprehensive use case descriptions

Table 2: Use Case Description for Login and Sign Up

Use case id	01
Use case name	Verify NGO
Actors	Admin
Description	Admin reviews NGO documents and approves or rejects the NGO account.
Trigger	NGO submits verification documents..
Pre-condition	NGO must be registered and must have uploaded documents.
Postcondition	NGO becomes Verified or Rejected.
Normal flow	Admin opens NGO requests. Admin reviews documents. Admin verifies or rejects NGO.
Exceptions	Missing or invalid documents.

Table 3: Approve Campaign

Use case id	02
Use case name	Approve Campaign
Actors	Admin
Description	Admin checks campaigns created by NGOs and approves or rejects them.
Trigger	NGO submits a new campaign.
Pre-condition	NGO must be verified.
Postcondition	Campaign becomes Active or Rejected.

Table 4: Browse Campaigns

Use case id	03
Use case name	Browse Campaigns
Actors	Donors
Description	Donor views the list of active donation campaigns.
Trigger	Donor opens campaigns page.
Pre-condition	Donors must be logged in.
Postcondition	Campaign list is displayed.
Normal flow	Donor opens campaign list. The system shows all active campaigns.
Exceptions	No campaigns available.

Table 5: View Campaign Details

Use case id	04
Use case name	View Campaign Details
Actors	Donor
Description	Donor views full details of a specific campaign.
Trigger	Donor selects a campaign.
Pre-condition	Campaign must be approved.
Postcondition	Donor sees complete campaign info.
Normal flow	Donor selects a campaign. System shows description, goal, updates, and NGO info

Table 6: Donate to Campaign

Use case id	05
Use case name	Donate to Campaign
Actor	Donor
Description	Donor donates a selected amount to a campaign.
Trigger	Donor clicks “Donate.”
Precondition	Donor is logged in, campaign is active
Postcondition	Donation recorded and campaign total updated.
Exception	Payment failed.
Normal flow	Donor chooses amount. System processes donations. System updates collected amount.

Table 7: View Donation History

Use case id	06
Use case name	View Donation History
Actor	Donor
Description	Donor checks all previous donations.
Trigger	Donor opens donation history.
Precondition	Donor is logged in
Postcondition	Donation history is shown.
Exception	No donations found.

Normal flow	Donor opens history. The system fetches and displays records.
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Table 8: Chatbot Assistance

Use case id	07
Use case name	Chatbot Assistance
Actors	Donor
Description	Donor asks the chatbot for campaign details and fund usage.
Trigger	Donor enters a query.
Precondition	Campaign must exist in the knowledge base.
Postcondition	Chatbot gives responses.
Normal flow	Donor asks a question. Chatbot checks knowledge base. Chatbot answers.
Exceptions	No matching data found.

Table 9: Register and Login

Use case id	08
Use case name	Register / Login
Actors	Donor / NGO
Description	The user creates an account or logs in.
Trigger	The user opens the login/register page.
Pre-condition	Users must enter valid credentials.
Postcondition	The user gets access to the dashboard.
Normal flow	User enters email/password. System verifies credentials. The user logs in.

Table 10: Upload Documents

Use case id	10
Use case name	Upload Documents
Actors	NGO
Description	NGO uploads verification documents for approval.
Trigger	NGO opens upload section.
Pre-condition	NGO must be registered.
Postcondition	Documents saved for admin review.
Normal flow	NGO selects documents. System uploads and stores them.

Table 11: Create Campaign

Use case id	11
Use case name	Create Campaign
Actors	NGO
Description	NGO creates a new donation campaign.
Trigger	NGO opens campaign creation page.
Pre-condition	NGO must be verified.
Postcondition	Campaign sent to admin for approval.
Normal flow	1. NGO enters campaign info. 2. System saves and sends to admin.
Exceptions	Missing required fields.

Table 12: Post Updates

Use case id	12
Use case name	Post Updates
Actors	NGO
Description	NGO posts progress updates for donors.
Trigger	NGO opens update section.
Pre-condition	Campaign must be approved.
Post-condition	Update visible to donors.
Normal Flow	NGO writes updates. System publishes updates.
Exception	Update content empty.

3.3 Functional Requirements

The functional requirements describe the core features and actions that the Hope Givers system must perform to meet the needs of donors, NGOs, and administrators. These requirements outline how users interact with the system, what operations they can perform, and how the platform should respond. They ensure that the system supports secure authentication, transparent donation management, NGO verification, campaign creation, chatbot assistance, and other essential tasks. By clearly defining these functions, the development process becomes organized, predictable, and aligned with the goals of providing a reliable and user-friendly donation platform.

These requirements include:

3.3.1 User Authentication and Authorization

Hope Givers requires a secure and reliable authentication system to ensure that only verified users can access the platform. Both donors and NGOs must create an account using valid credentials, and the system verifies their identity before granting access. Role-based authorization is applied, meaning each user is assigned permissions based on their role. Donors can browse campaigns and donate, while NGOs can upload documents and create campaigns.

The admin has the highest control and can verify NGOs and approve campaigns. This ensures that every user interacts only with the features that match their role, creating a safe and organized environment.

3.3.2 NGO Verification System

NGO verification system ensures that only authentic and trustworthy organizations can create fundraising campaigns on Hope Givers. When an NGO registers, it must upload verification documents such as identification records, registration certificates, or any proof of legitimacy.

These documents are then reviewed by the admin, who either approves or rejects the NGO based on authenticity. This verification step is essential because it prevents fraudulent organizations from creating campaigns and protects donors from scams. By verifying NGOs before they can post campaigns, Hope Givers maintains transparency, builds trust, and ensures that all donations go to real, verified causes.

3.3.3 Campaign Management

Campaign management is a core feature of Hope Givers that allows verified NGOs to create, manage, and update their donation campaigns. Once an NGO is approved, it can create a campaign by providing details such as campaign title, description, target amount, images, and purpose. These campaigns are then submitted to the admin for approval before becoming visible to donors. After approval, NGOs can post progress updates, upload new information, and keep donors informed about how funds are being used. This requirement ensures that campaigns remain organized, transparent, and regularly updated, giving donors confidence in the authenticity and progress of the causes they support.

3.3.4 Donation Processing System

The donation processing system allows donors to securely contribute to campaigns created by verified NGOs. When a donor selects a campaign, they can choose an amount and proceed to donate through a safe and encrypted process. Once the donation is completed, the system immediately updates the campaign's collected amount and records the transaction in the donor's history. This ensures full transparency and real-time tracking of funds. The secure handling of donations not only builds donor confidence but also ensures that every contribution is properly documented and reaches the intended campaign without risk of manipulation.

3.3.5 Chatbot Assistance

The chatbot assistance feature provides donors with quick and reliable information about campaigns, fund usage, and NGO activities. When donors ask a question, the chatbot retrieves verified details from the system's knowledge base and generates a clear and accurate response. This helps donors understand the purpose of a campaign, how previous donations were used, and what progress has been made. The chatbot ensures instant support, reduces confusion, and increases transparency by providing information based on trusted and approved data. Overall, this requirement improves user experience and builds donor confidence by offering immediate and verified guidance.

3.3.6 User Profile and History Management

User profile and history management allows each donor and NGO to maintain their personal information and track their past activities within the Hope Givers system. Donors can view and update their basic details, such as name, email, and password, and can also access their complete donation history to see when, where, and how much they contributed. NGOs can manage their organization information and keep their profiles updated with accurate documents and details. This requirement ensures a personalized experience, helps users stay organized, and allows both donors and NGOs to maintain a clear record of their interactions with the platform.

3.3.7 Search and Campaign Discovery

The search and campaign discovery feature helps donors quickly find the campaigns they are interested in. Donors can browse campaigns by category, urgency, NGO name, or keyword to easily locate causes they want to support. This system ensures that campaigns are organized and accessible, making it simple for users to explore different options without confusion. By improving visibility and navigation, this requirement enhances the donor experience and increases the chances of successful campaign engagement.

3.3.8 Admin Monitoring and Management

The admin monitoring and management feature ensures that the entire platform remains secure, organized, and transparent. The admin oversees all system activities, including reviewing NGO documents, approving campaigns, and monitoring donation records. The admin can remove suspicious content, verify authenticity, and maintain overall quality control. This requirement ensures that only trusted NGOs operate on the platform and that all campaigns follow proper

guidelines. By having full oversight, the admin helps protect donors from fraud and keeps the system running smoothly and responsibly.

3.3.9 Update Posting and Progress Tracking

The update posting and progress tracking feature allows NGOs to keep donors informed about how their contributions are being used. After a campaign is approved and donations begin, NGOs can post updates that include progress reports, images, or important milestones. These updates appear on the campaign page so donors can clearly see the impact of their contributions. This feature increases transparency and trust by ensuring that donors receive ongoing information, not just at the time of donation but throughout the entire campaign lifecycle. It also helps NGOs maintain credibility and strengthen communication with their supporters

3.4 Nonfunctional Requirements

Non-functional requirements describe the quality standards and performance expectations of the Hope Givers system. They focus on how well the system should operate rather than what functions it performs. These requirements ensure that the platform is secure, fast, reliable, easy to use, and capable of supporting future growth. They help maintain a smooth user experience and guarantee that the system performs efficiently under different conditions.

3.4.1 Performance

The system must provide fast and smooth performance, even when many users are active at the same time. Pages such as campaign lists, donation forms, and updates should load quickly without delays. The system should efficiently handle multiple donations, campaigns, and user requests to ensure a seamless experience for donors, NGOs, and admins.

3.4.2 Security

Security is a critical requirement for Hope Givers because it handles sensitive information such as donor details, NGO documents, and donation records. The system must protect all user data using strong encryption for passwords and secure storage methods for documents and transactions. Only authorized users should be able to access specific features through role-based permissions. All communication between the user and the system must be protected to prevent

unauthorized access, data leaks, or fraud. This ensures a safe and trustworthy donation environment for everyone.

3.4.3 Availability

This system must be always accessible, 24 hours a day, seven days a week.

3.4.4 Ease of Use

The Hope Givers system must be easy to use for all types of users, including donors, NGOs, and admins. The interface should be clean, simple, and intuitive so that users can easily browse campaigns, upload documents, donate, or post updates without confusion. Navigation should be straightforward, buttons and forms should be clear, and important features should be accessible with minimal effort. A user-friendly design improves the overall experience and reduces the learning curve for new users.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

The architecture of the Hope Givers system is based on a layered design that ensures smooth communication, security, and efficient data processing across all components. The Presentation Layer provides the web interface through which donors, NGOs, and administrators interact with the platform. Beneath it, the Business Logic Layer manages all core operations such as user authentication, NGO verification, campaign creation, donation processing, and chatbot response handling. These processes are supported by the Backend Services, including the database, file storage, payment handling, and knowledge base modules, which collectively ensure that data is stored securely and updated in real time. To protect users and sensitive information, the Security Layer applies role-based access control, encrypted data storage, and secure API communication. Together, these layers create a scalable, secure, and reliable architecture that supports transparent donation management and smooth system performance.

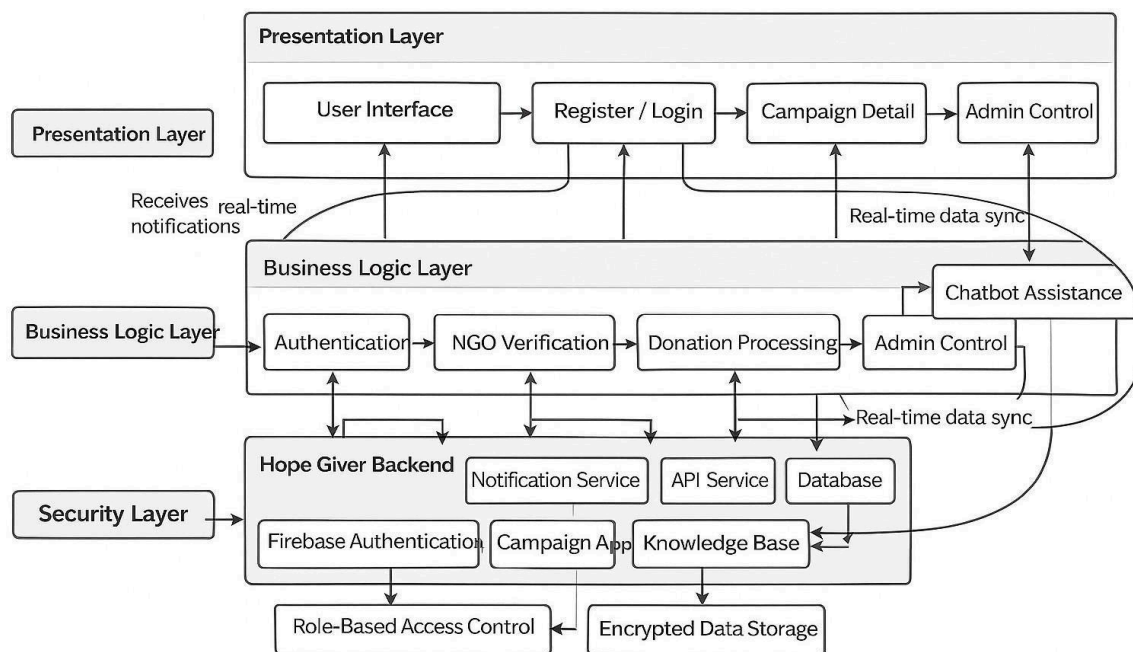


Figure 4.1: Hope Givers Architecture Diagram

4.2 Data Representation

The data representation of the Hope Givers system focuses on how information is organized and stored within the platform to support smooth operations. The system uses structured tables to manage user accounts, NGO details, campaigns, donations, and chatbot knowledge. Each table holds specific fields that ensure the data remains accurate, consistent, and easy to retrieve when needed. By using a well-organized data structure, the system can efficiently process user actions such as logging in, creating campaigns, posting updates, and recording donations. This structured representation also improves transparency and traceability, helping the platform maintain reliable and secure data management throughout the entire donation process.

4.3 Process Flow

The process flow of Hope Givers describes how different types of users Donors, NGOs, and Admins interact with the system from login to logout. The flow begins when any user registers and logs into the system. Once logged in, the system identifies the user's role and directs them to the appropriate dashboard.

- **User Flow**

The process starts when a new user registers on the platform. After successful registration, the user logs into the system and is shown a dashboard. At this point, the system evaluates the user's role. A decision is made to determine whether the user is a Donor, an NGO, or an Admin, and the process flow branches accordingly to the correct lane.

- **Donor Flow**

If the system identifies the user as a Donor, they enter the donor lane. Here, donors can browse available donation campaigns and view the details of any campaign they are interested in. After exploring a campaign, donors may choose to donate securely through the platform.

Once the donation is complete, donors can view their donation history to keep track of their contributions. In addition, they can check notifications for updates posted by NGOs

about the progress of the campaigns they supported. After completing their activities, the donor logs out to end their session.

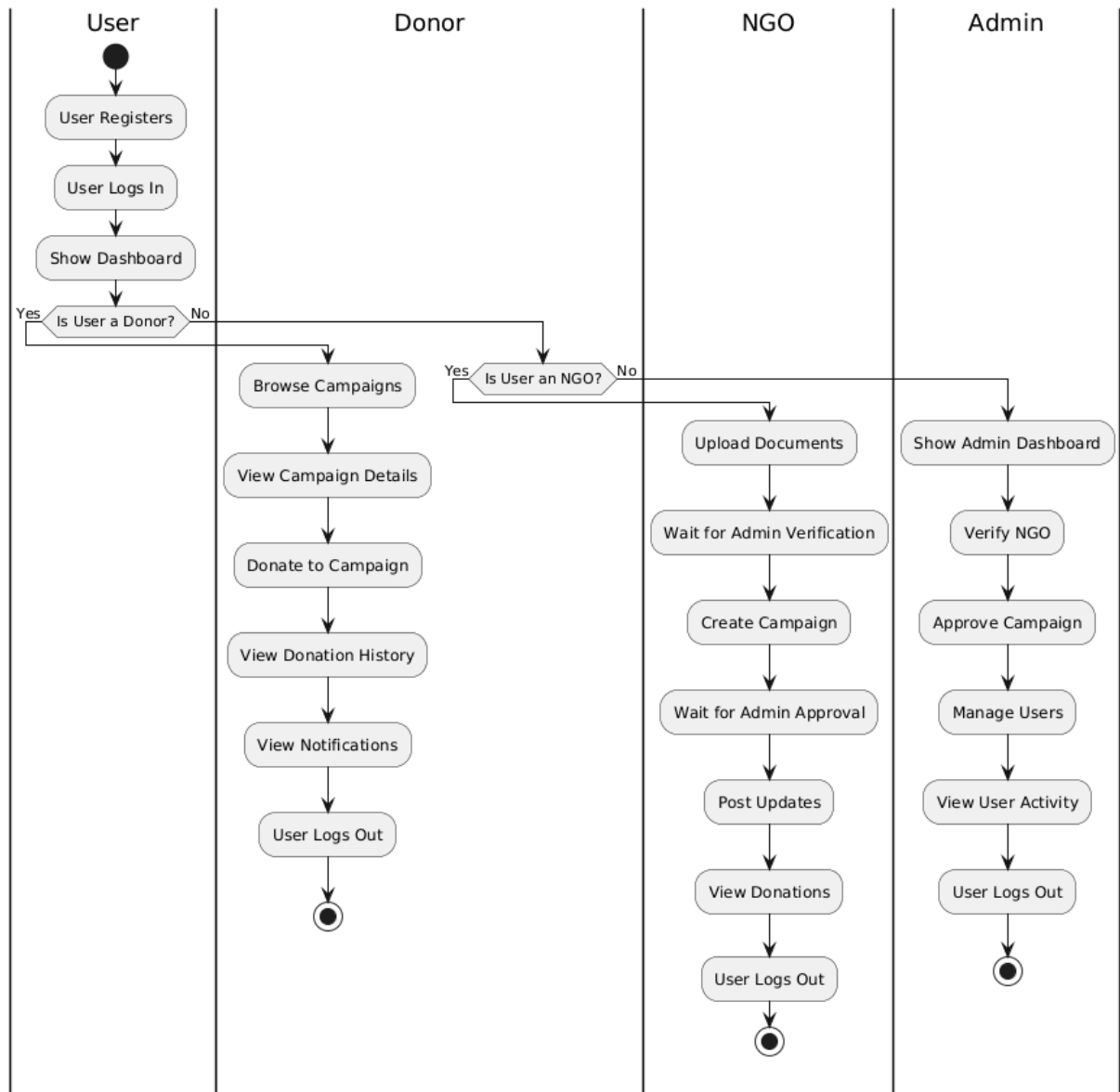


Figure 4.2: Hope Givers Process Flow Diagram

- **NGO Flow**

If the user is identified as an NGO, they enter the NGO lane. NGOs begin by uploading documents required for verification. After submitting these documents, they must wait for the admin to verify their identity and approve them as a legitimate organization. Once verified, NGOs can create new campaigns by adding details such as description, amount required, and

campaign goals. After creating a campaign, the NGO must again wait for admin approval to ensure authenticity and transparency.

When the campaign is approved, NGOs can post updates to keep donors informed, view incoming donations to track support, and maintain transparency. After completing tasks, the NGO logs out to end the process

- **Admin Flow**

If the system identifies the user as an Admin, they are taken to the admin dashboard. The admin's primary responsibility is to maintain the platform's reliability and transparency.

Admins verify the documents submitted by NGOs to ensure legitimacy. After verification, they review and approve campaigns created by NGOs. They also manage user accounts and monitor user activity to ensure the system is operating securely and correctly.

Once their administrative tasks are completed, the admin logs out to finish the session.

4.4 Sequence Diagram

These figures show how the system will work through sequence diagrams, task after task sequence wise. It gives a complete description of the working of Hope Givers.

Donor's Explanation

The donor sequence diagram explains how a donor interacts with the Hope Givers system from the moment they log in until they log out. After logging in, the donor accesses the donor dashboard, where they can browse available campaigns. When the donor selects a campaign, the system retrieves and displays the campaign details. If the donor chooses to donate, the system processes the donation securely and updates the campaign's collected amount. The donor can then view their donation history, which is fetched by the system based on their past contributions. Additionally, the donor can view notifications related to campaign updates posted by NGOs. Once all interactions are completed, the donor logs out, and the session ends. This sequence emphasizes a smooth and user-friendly donation experience.

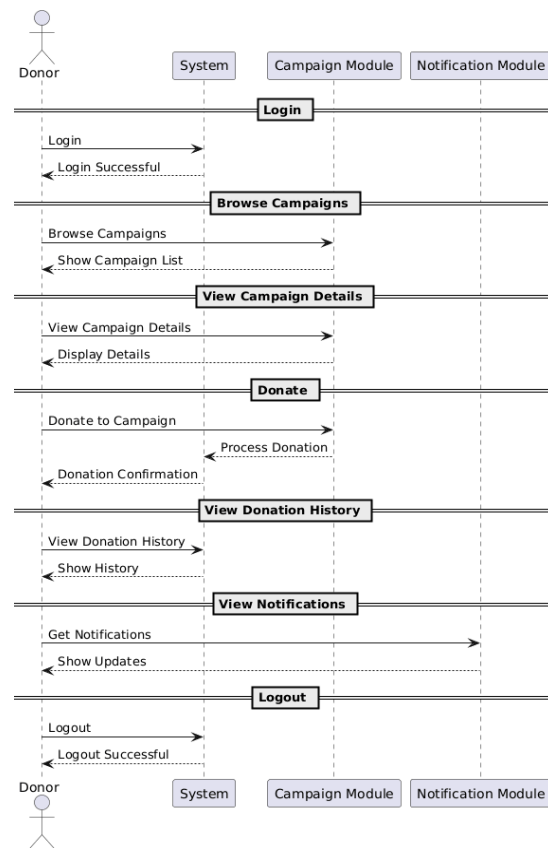


Figure 4.3: Hope Givers Donor Sequence Diagram

NGO Explanation

The NGO sequence diagram represents the complete workflow of an NGO inside the Hope Givers system. An NGO begins by logging in and accessing the NGO dashboard. The first major action is uploading verification documents, which the system receives and forwards to the admin for approval. After verification, the NGO can create new campaigns by providing all required details. The system then sends the campaign to the admin for review and approval. Once approved, NGOs can post updates related to campaign progress, which the system stores and displays to donors. NGOs can also view donations collected for their campaigns in real time. Finally, when their tasks are completed, NGOs log out of the system. This sequence highlights the transparency and accountability provided to donors through NGO actions.

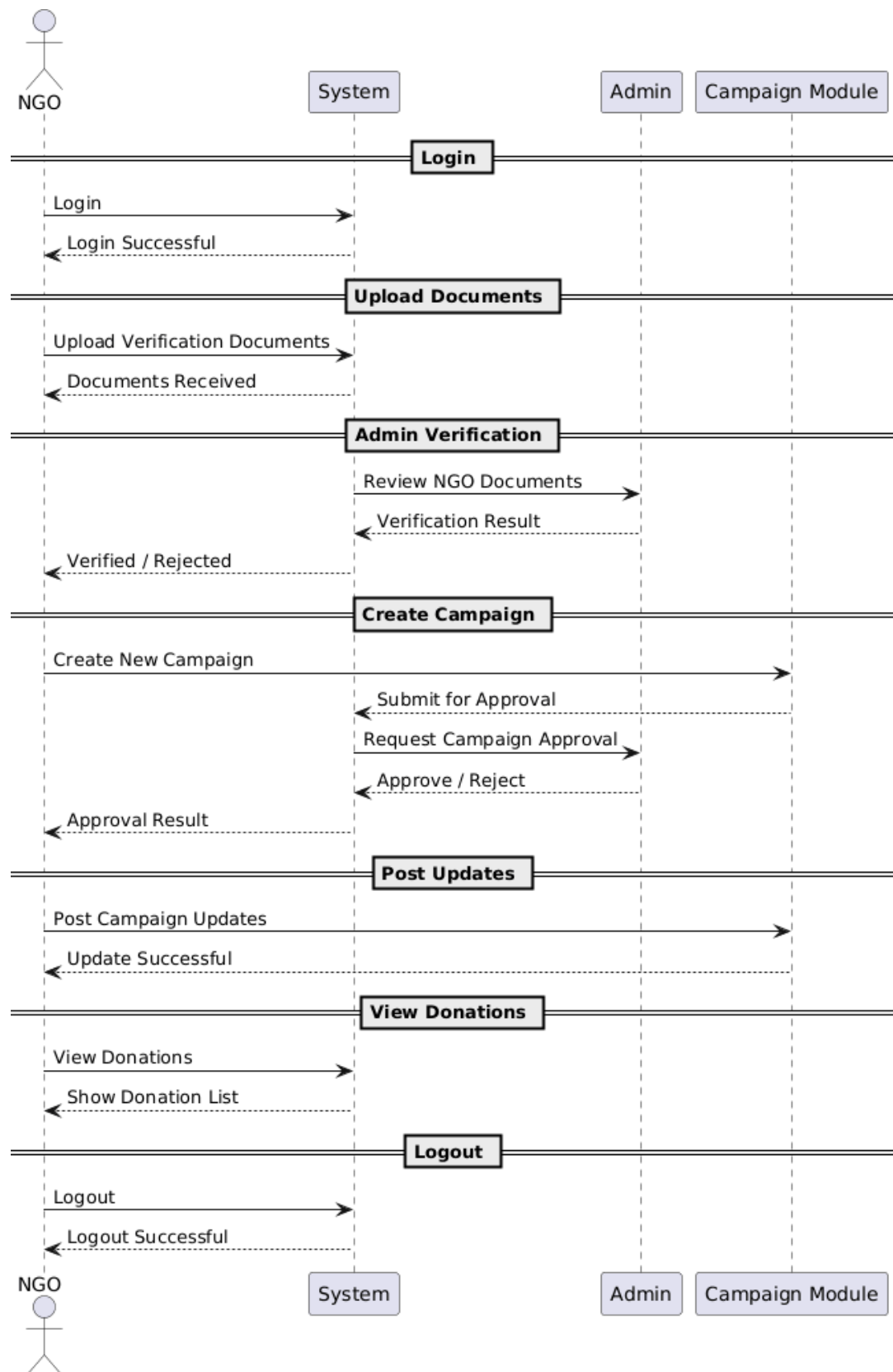


Figure 4.4: Hope Givers NGO Sequence Diagram

Admin Explanation

The admin sequence diagram illustrates how the admin manages and controls the Hope Givers system. After logging in, the admin gains access to the admin dashboard where they can verify NGO documents. When NGOs submit their documents, the system notifies the admin, who reviews and either approves or rejects them. Admins also review new campaigns submitted by NGOs and decide whether they should be approved based on authenticity and completeness. In addition, the admin manages all user accounts and can monitor platform activity through activity logs. These logs help track user actions for security and system monitoring. After completing all administrative duties, the admin logs out of the system. This sequence demonstrates how the admin maintains platform integrity, trust, and security.

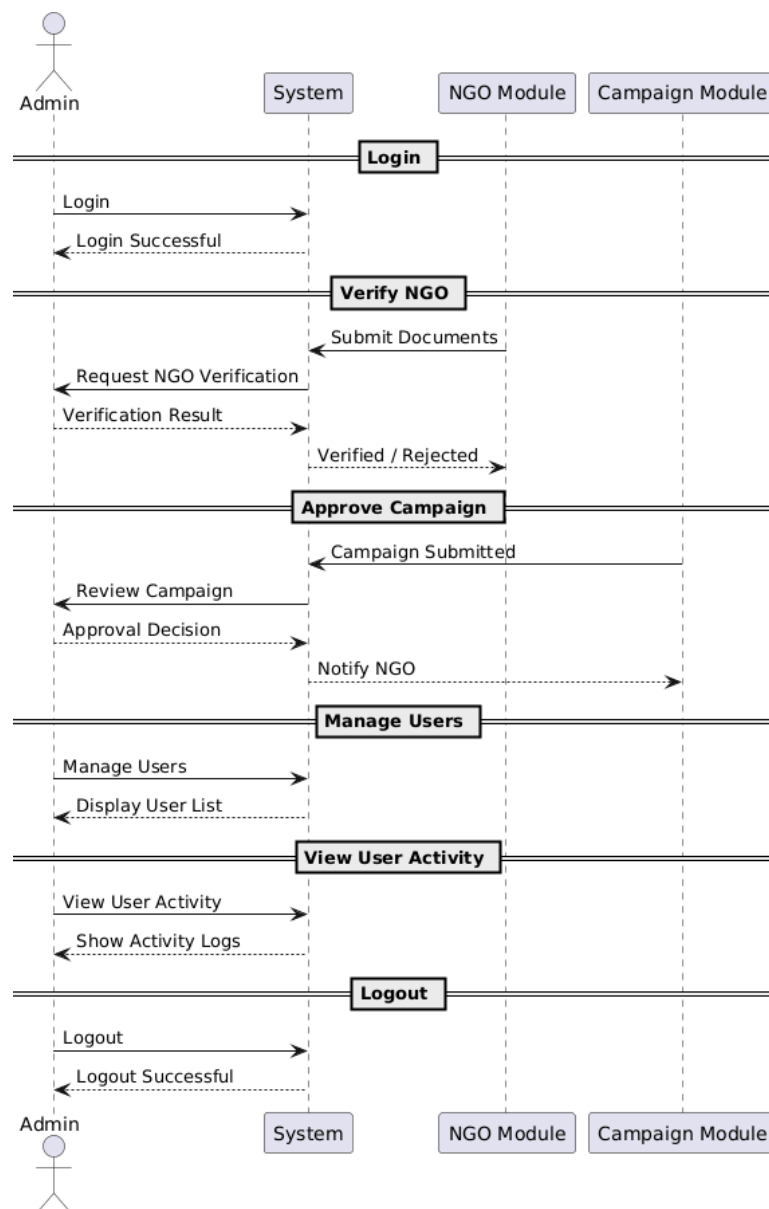


Figure 4.5: Hope Givers Admin Sequence diagram

4.5 Class Diagram

This class diagram is a graphical representation of the system's classes, their attributes, and the relationships between them. It elaborates the structure of the Hope Givers and shows how the different classes in the system are related to each other.

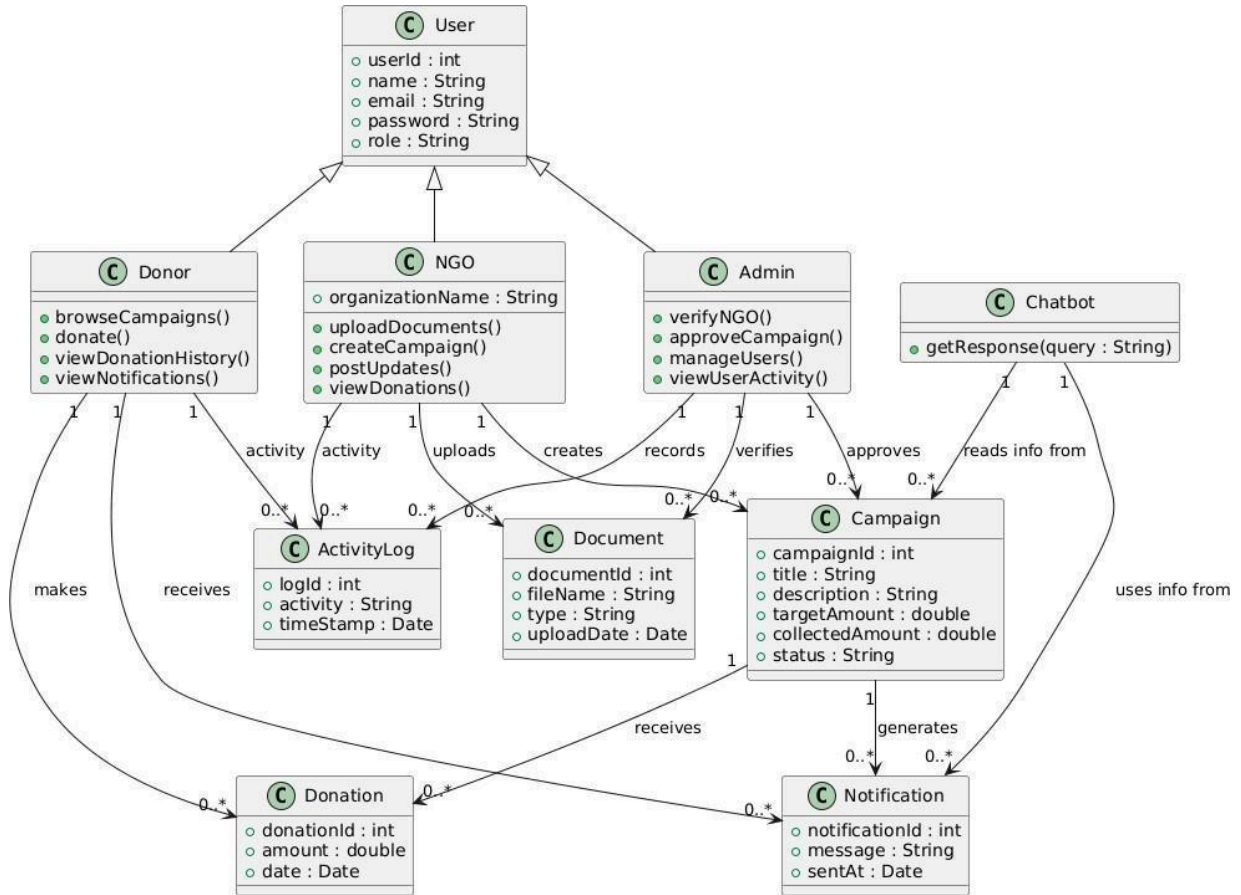


Figure 4.6: Hope Givers Class Diagram

4.5.1 Components:

User:

The User class represents the base entity for all system users. It stores essential information such as user ID, name, email, password, and role. This class provides the fundamental identity structure upon which other specialized user types are built. Donor, NGO, and Admin inherit these attributes and extend them with their own functionalities.

Donor:

The Donor class inherits from User and represents individuals who browse and donate to campaigns. The donor can:

- Browse available campaigns
- Donate to selected campaigns
- View donation history
- View notifications related to campaign updates

This class focuses on the donor-side operations of the Hope Givers platform.

NGO:

The NGO class extends the User class and represents verified organizations. NGOs can upload required verification documents, create new donation campaigns, post progress updates on campaigns, and view donations received for their campaigns.

This class supports the campaign management and transparency features of the system.

Admin:

The Admin class extends User and handles system governance. Admin responsibilities include:

verifying documents uploaded by NGOs, approving or rejecting NGO campaigns, managing all platform users, and viewing user activity logs.

The admin ensures security, authenticity, and proper functioning of the system.

Document:

The Document class stores all verification documents uploaded by NGOs. Each document includes an ID, file name, type, and upload date. This class enables the admin to verify NGO authenticity.

Campaign:

The Campaign class represents fundraising campaigns created by NGOs. It stores details such as campaign ID, title, description, target amount, collected amount, and status. Campaigns receive donations and generate notifications for donors.

Donation:

The Donation class represents a financial contribution made by a donor. Each donation includes an ID, amount, and date. Donations link donors to campaigns.

Notification:

The Notification class stores messages sent to donors regarding campaign updates, milestones, and general alerts. Each notification includes an ID, message, and timestamp.

Chatbot:

The Chatbot class is responsible for responding to donor queries using stored campaign and notification data. It uses system information to provide clear and automated responses.

Activity Log:

The ActivityLog class records user actions across the system. It includes a log ID, activity type, and timestamp. Activity logs help the admin track and monitor system behavior for security and auditing.

4.5.2 Relationships:

User Inheritance Relationship: The User class is the parent of Donor, NGO, and Admin. All three subclasses inherit common attributes and extend specialized functionalities.

NGO → Document: An NGO uploads multiple Documents. This relationship is 1 to 0..* and represents the document verification process.

NGO → Campaign: An NGO creates multiple Campaigns. This relationship is 1 to 0..* and represents campaign ownership.

Campaign → Donation: A Campaign can receive multiple Donations. Donors contribute to campaigns through this 1 to 0..* relationship.

Donor → Donation: A Donor can make multiple Donations. This relationship links donors to their donation history.

Campaign → Notification: A Campaign generates multiple Notifications for donors. This relationship reflects update and alert generation.

Donor → Notification: A Donor receives multiple Notifications from campaigns they support. Admin → Document: An Admin verifies multiple Documents submitted by NGOs.

Admin → Campaign: An Admin approves multiple Campaigns created by NGOs.

Chatbot → Campaign: The Chatbot reads information from campaigns to answer donor questions.

Chatbot → Notification: The Chatbot uses notification data to provide updated responses.

Admin → ActivityLog: Admins record actions in the system, such as approvals or verifications. Donor → ActivityLog: Donor actions (like donating or browsing) are stored in ActivityLog.

NGO → ActivityLog: NGO actions (campaign creation, updates) are also logged for transparency.

4.5.3 Summary:

The class diagram shows the main structure of the Hope Givers system. It begins with the User class, which is inherited by Donor, NGO, and Admin to represent different types of users. Supporting classes such as Campaign, Donation, Document, Notification, Chatbot, and ActivityLog handle core system operations. The relationships between these classes show how donors donate, NGOs create campaigns, admins verify activities, and the system maintains logs and updates. Overall, the diagram explains how all system components interact to support donation management.

Chapter 5

Implementation

5 Implementation

The implementation of the Hope Givers system focuses on converting the planned design into functional modules that support NGO verification, campaign creation, secure donations, chatbot assistance, and user interaction. Each feature was developed using a modular and layered approach to ensure smooth workflow and easy maintenance.

5.1 Algorithm

5.1.1.1 User Authentication

Function: authenticateUser(email, password)

Fetch users by email.

If found → match password hash.

If correct → return user; else → return error.

5.1.1.2 NGO Document Verification

Function: verifyNGO(user, documents)

Check if the user is an NGO.

Validate and store documents.

Mark status as “Pending Verification”.

5.1.1.3 Campaign Creation Function:

createCampaign(ngo, details)

Check if the NGO is verified.

Validate details and store campaign.

Set status as “Pending Approval”.

5.1.1.4 Donation Processing

Function: donate(donor, campaign, amount)

Ensure campaign is approved.

Process payment and record donation.

Update campaign collectedAmount.

5.1.1.5 Post Campaign Updates

Function: postUpdate(ngo, campaign, update)

- Verify NGO owns campaign.
- Save, update and notify donors.

5.1.1.6 Chatbot Interaction

Function: askChatbot(user,query)

- Check login status.
- Search knowledge base and return response.

Table 12: External APIs

Name of API	Description of APIs	Purpose of Usage	Function
Cloud Storage API	Store files online	File Service	NGO documents & images
Chatbot/NLP API	Auto-response system	Answer donor queries	Chatbot Module

5.2 User interface

The user interface of the Hope Givers system is designed to be simple, user-friendly, and responsive, ensuring a smooth experience for donors, NGOs, and administrators. Each screen is structured clearly to help users perform their tasks easily such as browsing campaigns, uploading documents, donating, or managing approvals.

Below are the main UI components of the system:

Figure 5.1: The Login Screen allows donors, NGOs, and admins to enter their email and password to securely access the system.

Figure 5.2: The Signup Screen enables new users to create an account by entering basic details and selecting their role (Donor, NGO, Admin).

Figure 5.3: The Feed Screen displays the main content area where users can explore active campaigns, updates, and featured causes. It provides quick navigation to different sections, helping users stay engaged with ongoing donation activities.

Figure 5.4: The Campaign List Screen displays a collection of available donation campaigns with their images, short descriptions, and target amounts to help donors explore and support causes.

Figure 5.5: The NGO Dashboard gives NGOs access to important functions such as uploading verification documents, creating campaigns, posting updates, and viewing donations received.

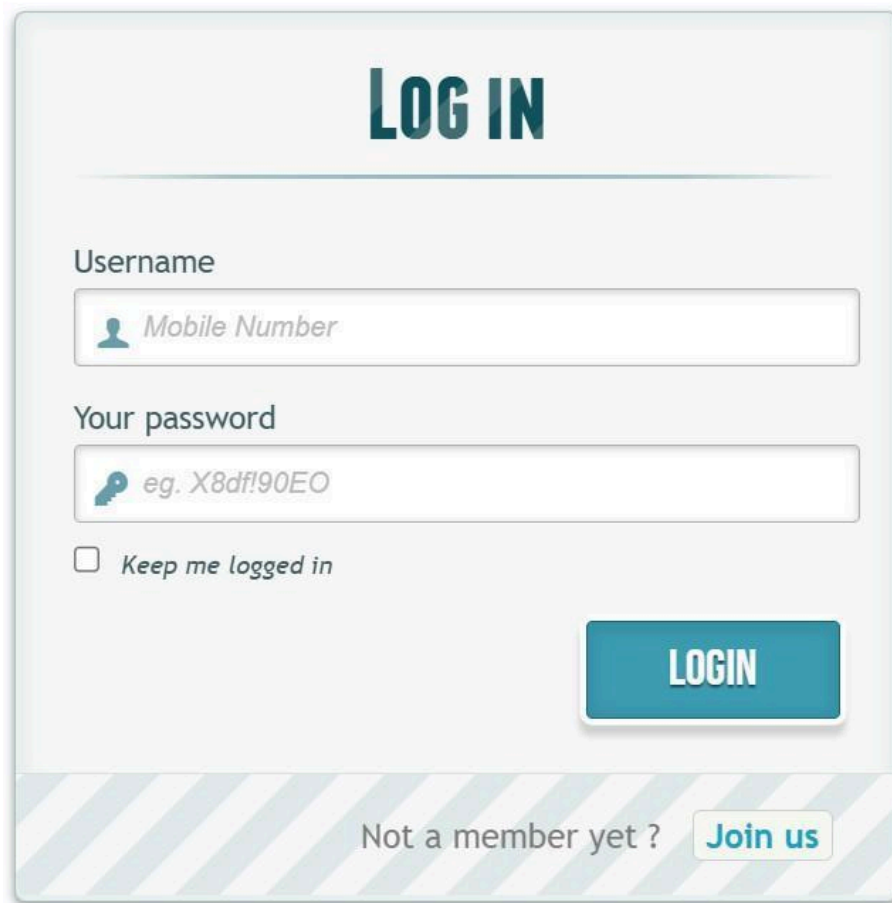
Figure 5.6: The Admin Dashboard allows administrators to verify NGO documents, approve campaigns, manage users, and monitor activity logs to maintain platform integrity.

Figure 5.7: The Donation Screen provides a secure interface for donors to enter the donation amount, choose a payment method, and complete the transaction.

Figure 5.8: The Chatbot Interface allows users to type queries and receive automated answers related to campaigns, donation usage, and NGO activities.

Figure 5.9: The Donor Dashboard provides access to all active donation campaigns, recommended causes, and navigation to key features like history, notifications, and chatbot.

HOPE GIVERS USERS' ACCESS POINT



The login screen features a light gray background with a white login form. At the top, the text "LOG IN" is displayed in a large, bold, dark blue font. Below this, the "Username" label is positioned above a text input field. Inside the field, there is a blue person icon and the placeholder text "Mobile Number". The "Your password" label is positioned above another text input field. Inside this field, there is a blue key icon and the placeholder text "eg. X8df!90EO". Below the password field, there is a checkbox with the label "Keep me logged in". To the right of these fields is a blue "LOGIN" button with white text. At the bottom of the form, there is a link "Not a member yet ?" followed by a "Join us" button with a blue border and blue text. The bottom of the form has a light blue and white diagonal striped background.

Figure 5.1: Hope Givers Log-In Screen

The login screen is the primary access point for all Hope Givers users, including donors, NGOs, and administrators. It provides fields for entering a username or mobile number and a secure password. A “Keep me logged in” option allows returning users to stay authenticated for convenience. The screen also includes a clear call-to-action button for logging in and an option for new users to navigate to the signup page. The interface is simple, clean, and focused on ensuring a smooth and secure user authentication experience.

HOPE GIVERS USERS' ACCESS POINT

SIGN UP

First Name

Middle Name

Last Name

Gender

Date Of Birth

Residence

Place of Birth

Ministry

Your email

Mobile Number

Password

SIGN UP

Figure 5.2: Hope Givers Sign-Up Screen

The signup screen allows new users to create an account on the Hope Givers platform. It includes fields for entering personal details such as first name, middle name, last name, gender, date of birth, residence, and place of birth. Users must also provide their email, mobile number, and choose a password for secure access. Dropdown menus and text fields are used to ensure accurate data entry. Once the form is completed, the user can tap the “Sign Up” button to

register. This screen ensures all required user information is collected in a clean and organized layout.

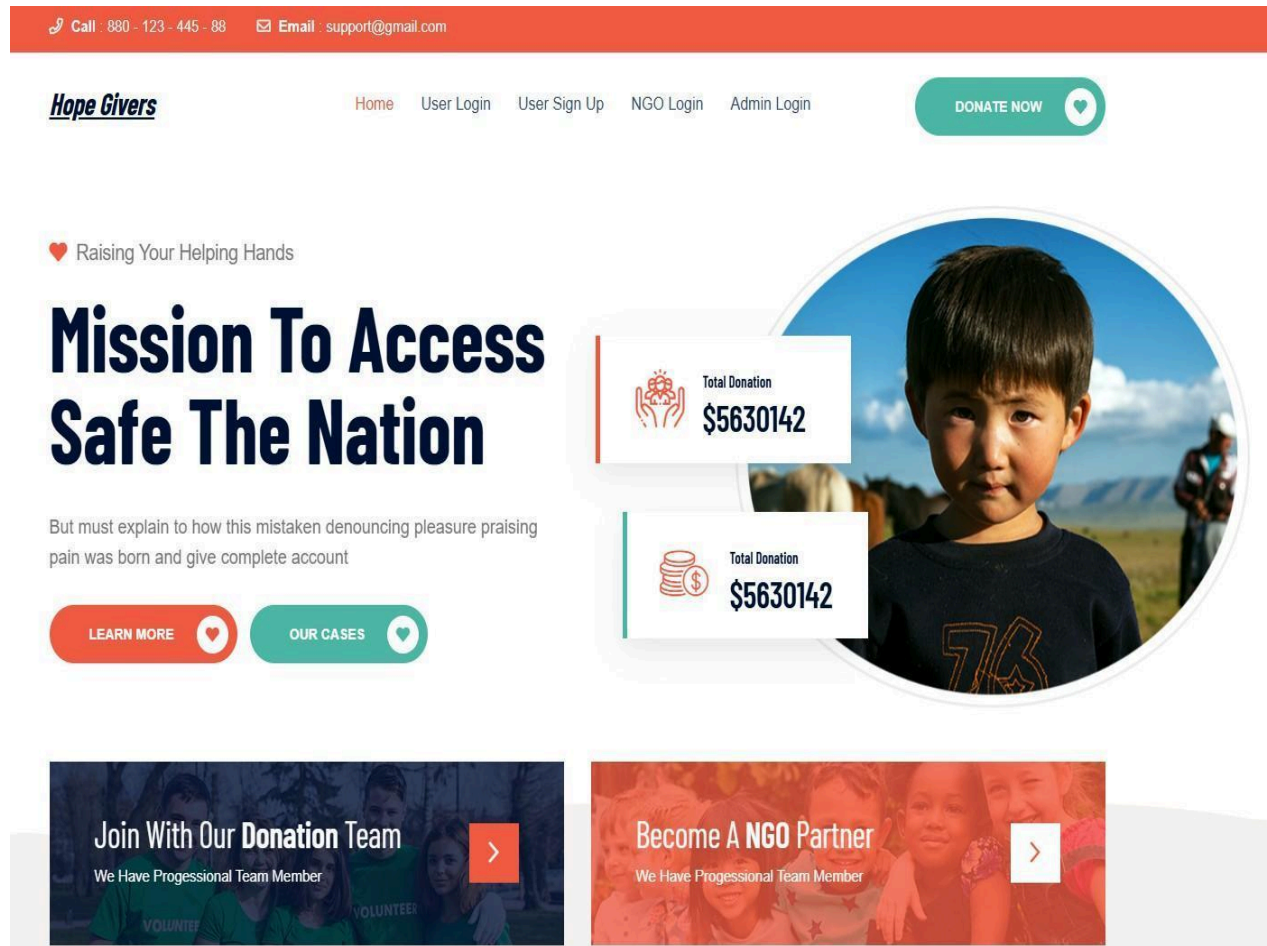


Figure 5.3: Hope Givers Feed Screen

The feed screen serves as the main homepage where users can explore donation campaigns, view total donations, and access platform features such as NGO login, Admin login, and “Donate Now.” It highlights key causes with visuals and call-to-action buttons, guiding users toward active campaigns and engagement options.

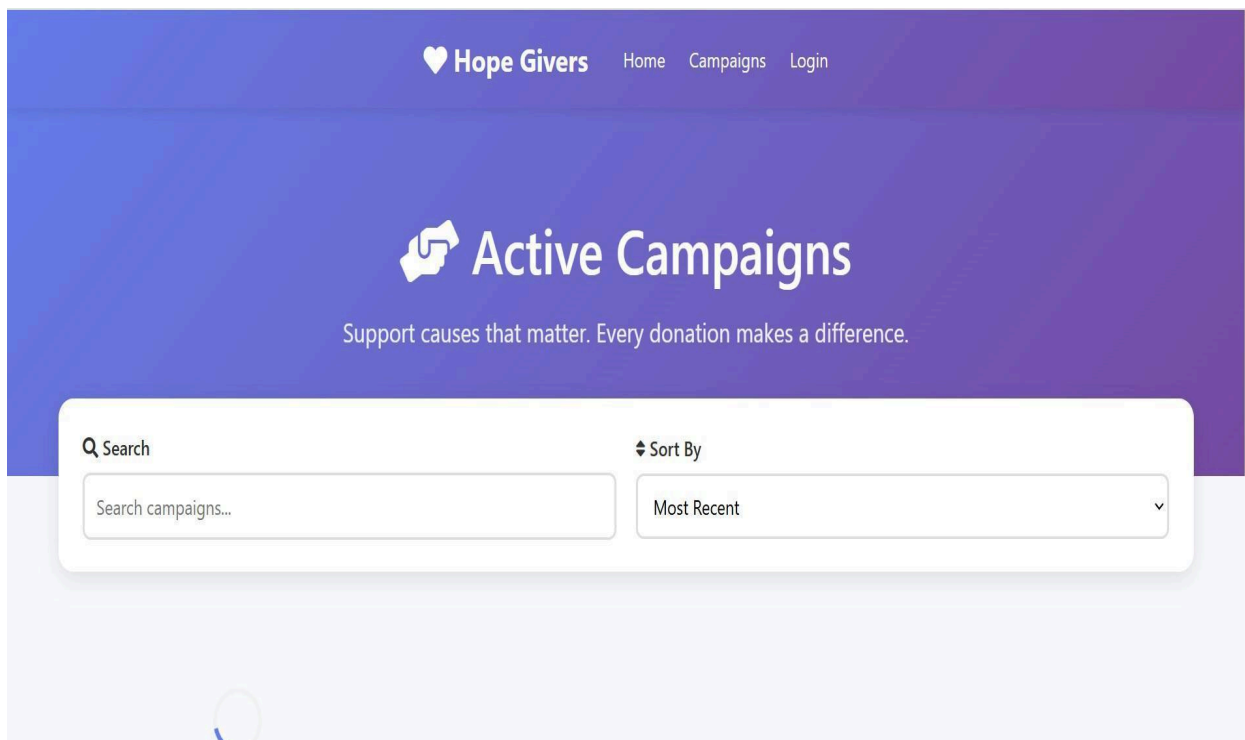


Figure 5.4: Hope Givers Campaign List Screen

The campaign list screen provides users with a structured view of all available donation campaigns. Each campaign card displays an image, title, short description, and target amount, helping donors quickly understand the cause. The list format makes it easy to scroll and explore multiple campaigns at once. This screen helps users compare and select the campaigns they wish to support.

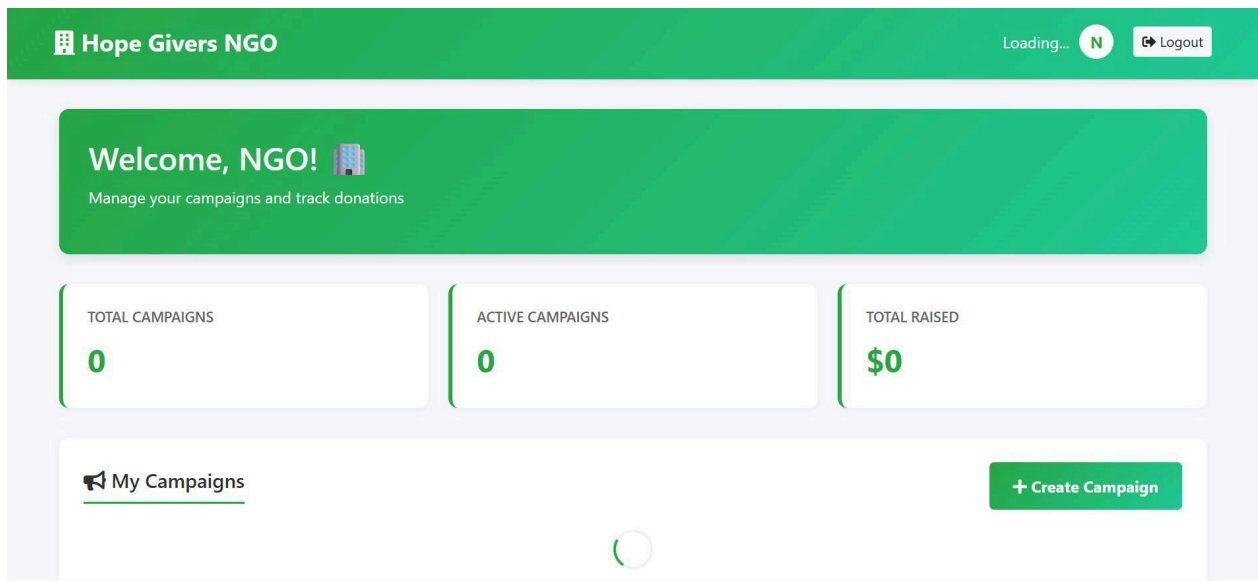


Figure 5.5: Hope Givers NGO Dashboard Screen

The NGO dashboard gives verified NGOs access to essential tools for managing their fundraising activities. From this screen, NGOs can upload verification documents, create new campaigns, view donations received, and post campaign updates. A clean menu layout ensures NGOs can easily navigate tasks without confusion. This dashboard provides a centralized workspace for NGOs to maintain transparency and manage campaigns efficiently.

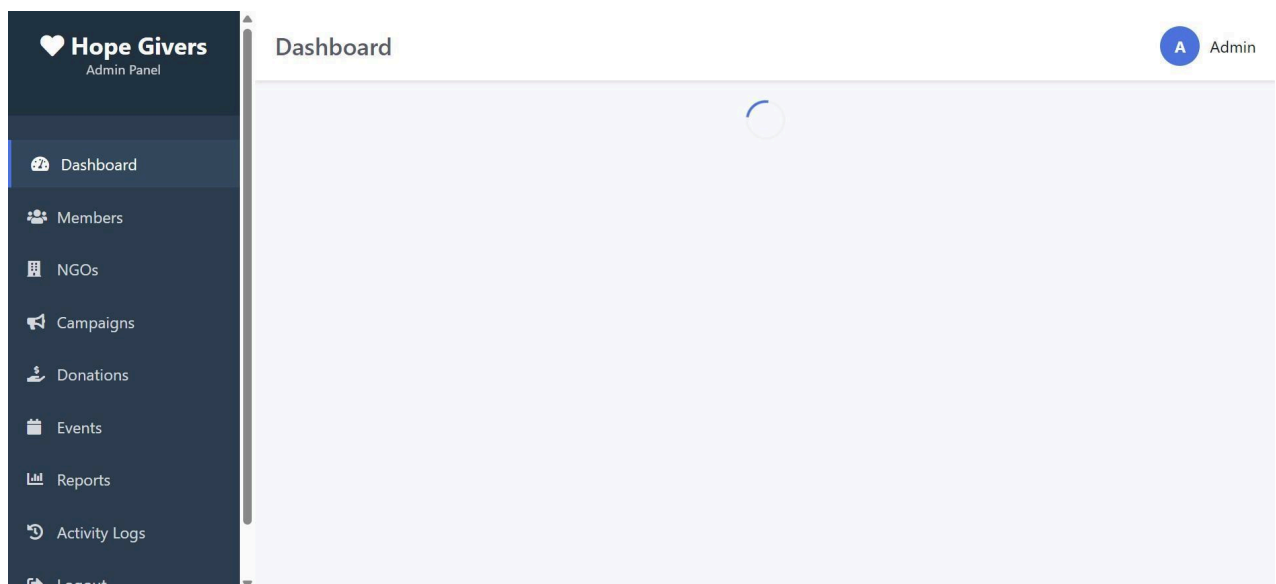


Figure 5.6: Hope Givers Admin Dashboard Screen

The admin dashboard provides administrators with complete control over platform operations. It allows admins to review NGO verification documents, approve or reject campaigns, manage user accounts, and monitor system activity. The layout organizes all administrative functions clearly to support quick decision-making. This screen ensures that the system remains secure, authentic, and well-regulated.

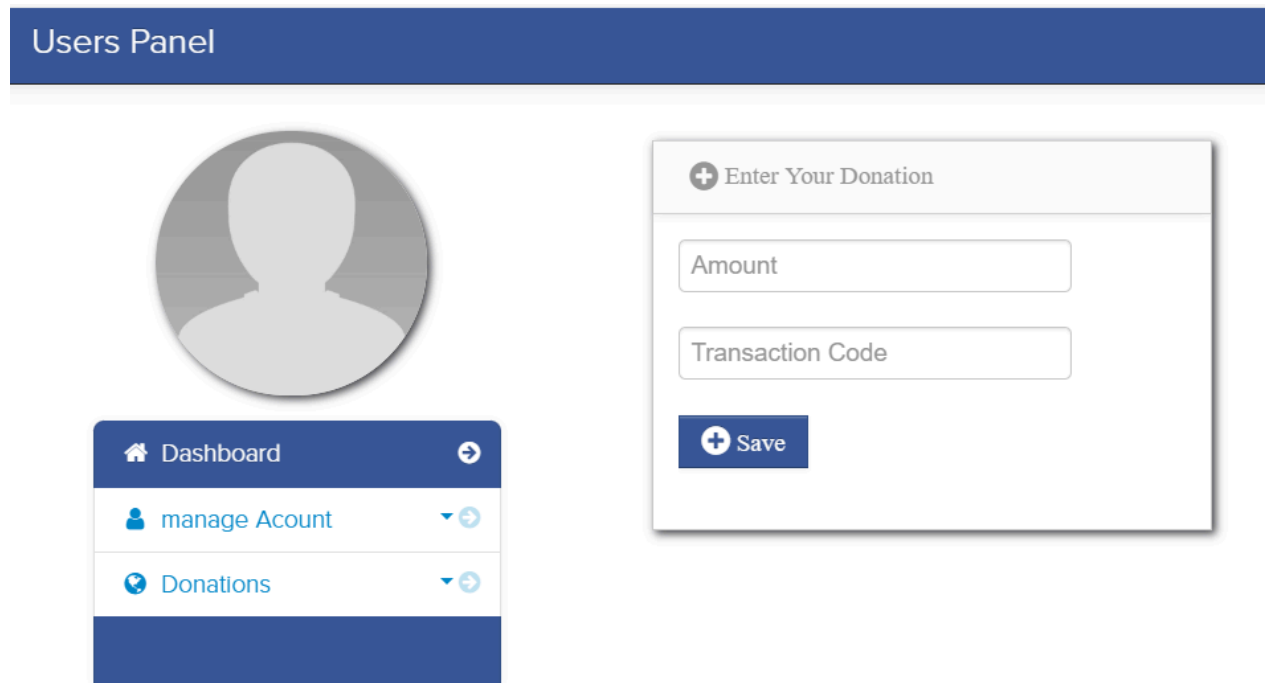


Figure 5.7: Hope Givers Donation Screen

The donation screen allows donors to securely contribute to a selected campaign. Users can enter their donation amount, choose a payment method, and proceed with the transaction using clear and secure input fields. The screen displays campaign details to reassure donors that their funds are being directed to the right cause. This interface ensures a simple, smooth, and secure donation experience.

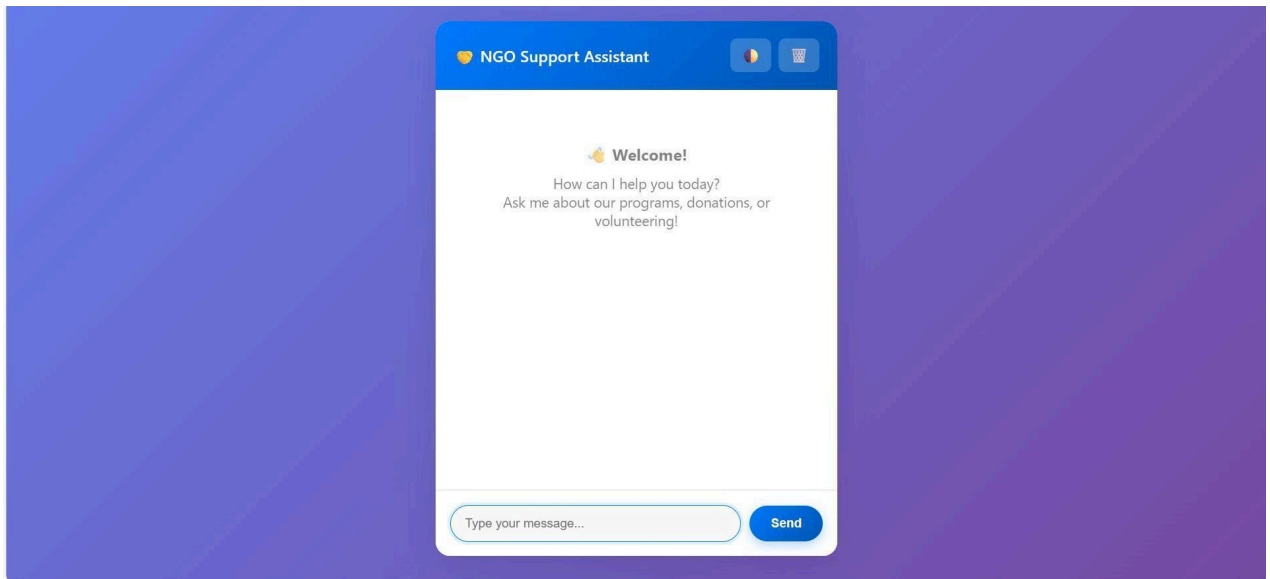


Figure 5.8: Hope Givers Chatbot Interface

The chatbot interface provides users with an easy way to ask questions and receive instant automated responses regarding campaigns, donations, or NGO activities. It includes a text box for entering queries and a message area displaying responses. This screen enhances user engagement by offering quick assistance without navigating multiple pages. It makes the system more interactive and user-friendly.

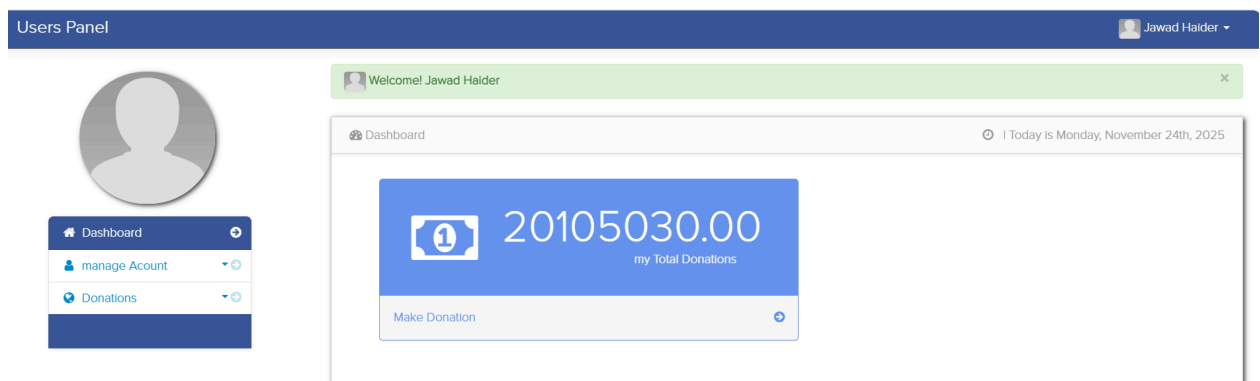


Figure 5.9: Hope Givers Donor Dashboard Screen

The donor dashboard presents a personalized view of campaigns tailored for each donor. It shows active campaigns, recommended causes, recent updates, and access to donation history and notifications. Quick navigation buttons help donors explore campaigns or check their past contributions effortlessly. This dashboard ensures donors can manage and track their charitable activities in a streamlined manner.

Chapter 6

Testing and Evaluation

6 Testing and Evaluation

The testing and evaluation phase ensures that the Hope Givers system works correctly, meets functional requirements, and provides a smooth and secure user experience. During this phase, each module of the system such as user authentication, NGO verification, campaign management, donation processing, chatbot responses, and notification handling was tested individually and then evaluated as part of the integrated system. The purpose of this chapter is to verify system behavior, identify and fix defects, and confirm that the system performs as expected under different scenarios. The evaluation also checks whether the system aligns with user needs, maintains data accuracy, and functions reliably during real usage.

6.1 Manual Testing

Manual testing for the Hope Givers system involves testers manually interacting with the platform's features to ensure everything functions as expected. Testers act like real donors, NGOs, and admins to verify that login, document uploads, campaign creation, donations, chatbot responses, and notification features behave correctly. This method helps identify UI issues, navigation problems, and functional errors. Since manual testing does not analyze internal code behavior, some deeper or hidden issues may remain undiscovered, but it is still essential for validating core user flows and overall usability.

6.1.1 System Testing

System testing checks whether the entire Hope Givers platform works seamlessly after all modules are integrated. It ensures the system meets its functional requirements, handles data properly, and provides expected results under normal usage. This includes verifying that donor interactions, NGO campaign management, admin approvals, donations, and chatbot responses operate smoothly from end to end. System testing helps uncover issues that may only appear when different modules work together and ensures the platform is ready for deployment.

6.1.2 Unit Testing

Unit testing in the Hope Givers system focuses on evaluating individual components separately before integration. Each module such as login authentication, document upload, campaign creation, donation processing, and chatbot response generation is tested in

isolation to detect logic errors early. For example, unit tests verify correct input formats (email, password strength, amount validation), check if campaign forms accept valid data, and ensure donation calculations update correctly. Unit testing allows developers to repeatedly re-run checks whenever code is updated, ensuring each piece of the system remains reliable before combining modules for integration testing.

Unit Testing 1: Login as an Admin

Testing Objective: To ensure the login form and signup form are working correctly

Table 13: Login as an Admin

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Admin Login with valid credentials	Username: naveedhafeez01@gmail.com Password: Admin@123	Admin successfully logs into the system	Pass
UT-002	Admin Login with invalid credentials	Username: wrongadmin@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass
UT-003	Admin Login with empty fields	Username: Password:	Prompt: "Please enter email and password"	Pass

Unit Testing 2: Login as NGO

Testing Objective: To ensure the login and signup forms are functioning correctly for NGO in the Hope Givers.

Table 14: Login as NGO

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
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UT-001	NGO login with valid credentials	Username: danishzafar@gmail.com Password: danish1123 Username: alihaider@gmail.com Password: hello22	NGO successfully logins	Pass
UT-002	NGO login with invalid credentials	Username: ward@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass

Testing Objective: To ensure the login and signup forms are functioning correctly for user in the Hope Givers.

Table 15: Login as a Donor

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Donor login with valid credentials	Username: alinawaz1@gmail.com Password: alinawaz123	Donor successfully logs in to the system	Pass
UT-002	Donor login with invalid credentials	Username: ward@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass

Unit Testing 4: Logout as an Admin

Testing Objective: To ensure that the Parent successfully log out from the system

Table 16: Logout as an Admin

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Click the logout button and logout	Logout	Admin Successfully exit from the system	Pass

Unit Testing 5: Logout as NGO

Testing Objective: To ensure that the NGO successfully logged out from the system

Table 17: Logout as a NGO

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Click the logout button and logout	Logout	NGO Successfully exit from the system	Pass

Unit Testing 6: Logout as a Donor

Testing Objective: To ensure that the User successfully logged out.

Table 18: Logout as a Donor

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	Click the logout button and logout	Logout	Donor Successfully exit from the system	Pass

6.1.3 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the modules is tested. This is to ensure that the system produced meets the specifications and requirements.

Functional Testing 1: Login as Admin

Objective: To ensure that the correct dashboard and features for admin are displayed and

functional upon login.

Table 19: Login as an Admin

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	Admin login with valid credentials	Username: naveedhafeez@gmail.com Password: Admin@123	Admin Successfully logged in to the system	Pass
FT-002	Admin Login with invalid credentials	Username: wrongadmin@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass
FT-003	Approve NGO documents	NGO uploads: CNIC, Registration Proof	NGO status updated to Verified	Pass
FT-003	Approve NGO campaign	Campaign Details: Title, Amount, Description	Campaign status updated to Approved	Pass
FT-004	Manage users	Action: Enable/Disable specific Donor/NGO accounts	User accounts managed successfully	Pass

FT-005	Send notification	Notification Type: Campaign updates, Alerts	Notification sent to selected users	Pass
FT-006	Logout	Click the logout button and logout	Admin Successfully exit from the system	Pass

The functional testing table validates the major administrative features of the Hope Givers platform. Test Case FT-001 verifies that the admin can log in using valid credentials (naveedhafeez@gmail.com, Admin@123). FT-002 confirms the system correctly rejects invalid login attempts by displaying “Invalid credentials.” FT-003 tests the document verification process, confirming that the admin can successfully review and verify NGO-uploaded documents. FT-004 checks campaign approval functionality, ensuring campaigns submitted by NGOs are processed and updated to “Approved” status. FT-005 evaluates user management actions such as enabling or disabling donor/NGO accounts. FT-006 tests the admin’s ability to send notifications like updates and alerts to selected users. Finally, FT-007 confirms that the admin is successfully logged out and returned to the login screen. All tests passed, demonstrating that the admin panel functions reliably.

Functional Testing 2: Login as a NGO

Objective: To ensure that after login, all NGO features work properly, including document upload, campaign creation, and updates.

Table 20: Login as NGO

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	NGO login with valid credentials	Email: hopefoundation@gmail.com Password: Hope@123	NGO successfully logs into the system	Pass
FT-002	NGO Login with invalid credentials	Username: wrongpresi@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass

FT-002	Upload verification documents	CNIC, Registration Certificate, Address Proof	Documents uploaded and marked Pending Verification	Pass
FT-003	Create new campaign	Title, Goal Amount, Description, Image	Campaign created and sent for Admin Approval	Pass
FT-004	Post campaign update	Update text, Image (optional)	Update posted and visi	Pass
FT-005	View Donations	Click on “Donations” tab	All donations to NGO campaigns displayed	Pass
FT-006	Click the exit button and logout	Logout	NGO successfully logged out	Pass

The testing ensures that all NGO-related functionalities work as intended. FT-001 confirms successful login using valid credentials, while FT-002 validates that incorrect credentials are rejected. FT-003 verifies the document upload process, marking the NGO as "Pending Verification." FT-004 tests campaign creation, ensuring campaigns are submitted for admin approval. FT-005 validates posting campaign updates that appear to donors. FT-006 checks if NGOs can view donations received for their campaigns. FT-007 verifies proper logout behavior. All tests passed, confirming a functional NGO module.

Functional Testing 3: Login as Donor

Objective: To ensure that students can access society-related features and interact with them properly in the CUI Sphere app.

Table 21: Login as Donor

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	Donor Login with valid credentials	Email: donor123@gmail.com Password: Donor@123	Donor successfully logs in to dashboard	Pass
FT-002	Donor Login with invalid credentials	Username: wrongstu@gmail.com Password: wrong pass	Error message: "Invalid credentials"	Pass
FT-002	View campaign list	Navigate to "Campaigns"	Campaigns displayed correctly	Pass
FT-003	Donate to a campaign	Amount: 500 PKR, Payment Method	Donation processed successfully	Pass
FT-004	View donation history	Click "History" tab	All previous donations displayed	Pass
FT-005	View campaign updates	Open "Updates" tab	Latest updates from supported campaigns displayed	Pass

FT-006	Update profile	New profile picture, Name, Password	Profile successfully updated	Pass
FT-007	Logout from app	Click on "Logout" button	Donor successfully logs out	Pass

This table verifies critical donor features. FT-001 confirms the donor can log in successfully; FT-002 ensures incorrect logins show appropriate error messages. FT-003 checks the campaign list display. FT-004 verifies the donation process, ensuring payments are handled securely. FT-005 confirms donation history is shown correctly. FT-006 tests viewing campaign updates posted by NGOs. FT-007 validates the logout process. All tests passed, indicating the donor module functions effectively.

6.1.4 Integration Testing

Integration testing is a crucial phase in software development that focuses on combining individual units or modules and testing them as a group. This process helps identify defects in the interactions between modules, ensuring they communicate correctly, and function together as intended. By detecting interface issues, logical errors, and potential data mismatches, integration testing confirms that the application operates smoothly in a real-world environment. It ultimately enhances software reliability before proceeding to further testing stages.

Integration Testing 1: Login as Admin

Objective: To ensure that the admin panel integrates correctly with modules such as NGO verification, campaign approval, user management, and notifications.

Table 22: Login as an Admin

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	Admin logs in and verifies NGO documents	Login Module, NGO Verification Module	Admin successfully logs in and verifies NGO documents	Pass

IT-002	Admin approves NGO campaign	Campaign Approval Module, Notification Module	Campaign is marked as Approved, notification sent to NGO	Pass
IT-003	Admin edits user information	User Management Module, Database Module	User info updated and reflected in database	Pass
IT-004	Admin enables/disables user accounts	Account Control Module, Authentication Module	Account status updated and affects login permissions	Pass
IT-005	Admin logs out	Admin Dashboard, Logout Module	Admin is logged out successfully	Pass

Integration testing ensures that the admin module works smoothly with other connected modules. In IT-001, the admin logs in and verifies NGO documents, confirming integration between the login and verification modules. IT-002 checks campaign approval and notification delivery, confirming correct interaction between approval and notification modules. IT-003 validates that admin updates to user profiles reflected in the database. IT-004 tests enabling/disabling user accounts, ensuring these changes affect login permissions via the authentication module. IT-005 confirms that the logout procedure integrates correctly with the dashboard and session modules. All tests passed, confirming correct system behavior.

Integration Testing 2: Login as NGO

Objective: To ensure that the correct modules load for NGOs and that all NGO functionalities integrate properly.

Table 23: Login as NGO

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	NGO logs in and creates a campaign	Login Module, Campaign Creation Module	Campaign created and sent to admin for approval	Pass
IT-002	NGO uploads verification documents	Document Upload Module, File Storage Module	Documents uploaded and status updated	Pass
IT-003	NGO posts campaign update	Update Module, Notification Module	Update posted and notifications sent to donors	Pass
IT-004	NGO views donation list	Donation Module, Database Retrieval Module	Donation list fetched and displayed correctly	Pass
IT-005	NGO Logs out	NGO Dashboard, Logout Module	NGO logged out successfully	Pass

Integration testing verifies how NGO modules interact with other components. IT-001 confirms that NGOs can log in and create campaigns, integrating the login and campaign creation modules. IT-002 validates document upload, ensuring files are saved correctly through the storage module. IT-003 checks posting updates and triggering notifications for donors. IT-004

ensures donation data retrieval integrates correctly with the database. IT-005 verifies that the logout process works seamlessly. All tests passed, showing smooth NGO workflow integration.

Integration Testing 3: Login as Donor

Objective: To ensure donor features integrate correctly, including browsing, donating, receiving updates, and profile operations.

Table 24: Login as a Donor

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	Donor logs in and lands on dashboard	Login Module, Navigation Module	Donor lands on campaign dashboard	Pass
IT-002	Donor views and supports campaign	Campaign Module, Donation Module	Donation recorded and campaign updated	Pass
IT-003	Donor views campaign updates	Update Module, Notification Module	Updates correctly displayed	Pass
IT-004	Donor views donation history	Donation History Module, Database Module	Donation history loaded successfully	Pass
IT-005	Donor logs out	Dashboard Module, Logout Module	Donor logged out and returned to login screen	Pass

Integration testing for donors ensures all donor-related modules work cohesively. IT-001 confirms proper redirection to the dashboard after login. IT-002 validates interactions between campaign browsing and the donation module. IT-003 ensures campaign updates integrate with notification and display modules. IT-004 checks retrieval of donation history from the database. IT-005 confirms a proper logout sequence. All tests passed, demonstrating the donor module integrates correctly with the system

6.2 Tools and Technologies

Table 25: Tools and Technologies for Proposed Projects

Tools And Technologies	Tools	Version	Rationale
	VScode	Latest	Used as the primary IDE for frontend and backend development.
	XAMPP	Latest	Provides Apache, PHP, and MySQL environment for running and testing the Hope Givers system.
	MySQL	8.0+	Stores users, NGOs, campaigns, donations, notifications, and system logs.
	MS PowerPoint	365	Presentation Creation
	MS Word	365	Project Documentation
	Technology	Version	Rationale
	HTML5 / CSS3 / JavaScript	Latest	Used for building the frontend interface and UI components.
	PHP	8.0+	Used to implement backend logic, sessions, database operations, and API interactions.
	Cloud Storage	Latest	Used to store NGO documents and campaign images securely.
	Firebase Authentication	Latest	User Login/Signup for Donor/admin

6.3 Summary

This chapter presented the different types of testing performed on the Hope Givers system, including manual testing, functional testing for each user role (Admin, NGO, and Donor), and integration testing to ensure modules work together smoothly. Each test case was executed using both valid and invalid scenarios to verify login, campaign creation, donations, document uploads, profile management, and update posting. The actual system behavior was compared to expected outcomes, and all test cases passed successfully, demonstrating that the Hope Givers system functions reliably and meets its intended requirements.

Chapter 7

Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The Hope Givers system has been designed to create a transparent, organized, and user-friendly donation platform that connects donors, NGOs, and the system administrator. The primary goal of this project was to build a digital environment where NGOs can easily submit their verification documents, create campaigns, share updates, and view donations; donors can browse campaigns, contribute securely, and stay informed; and the admin can verify NGOs, approve campaigns, and manage system-wide activities.

The system successfully simplifies the traditional, time-consuming donation process by shifting it into a fully digital platform. Instead of relying on manual verifications, physical meetings, or scattered social media posts, Hope Givers provides a centralized platform where all donation-related information is available in one place. Donors can now explore causes confidently, NGOs can showcase their work transparently, and administrators can maintain control and oversight with ease.

Another strength of this system is its clean, intuitive, and easy-to-navigate interface. Users of all roles—Donor, NGO, and Admin—can interact with the system without any technical barriers. Whether someone wants to donate to a cause, upload documents, approve campaigns, or view donation summaries, the interface ensures a smooth workflow.

Throughout the implementation and testing phases, major features such as login authentication, document verification, campaign creation, donation processing, chatbot interaction, notifications, and user management were thoroughly tested. All modules performed as expected, confirming the reliability and stability of the system.

In conclusion, the Hope Givers platform has achieved its goal of improving trust, transparency, and convenience in the donation process. It brings together donors, NGOs, and administrators in one integrated system, promoting accountability and making charitable giving more accessible and impactful.

7.2 Future Work

Although the Hope Givers system successfully fulfills its core objectives, there is significant potential for enhancement. The following improvements are planned for future versions of the platform:

7.2.1 Advanced Analytics Dashboard

Add analytic tools for NGOs and Admin to view donation trends, campaign performance, donor engagement, and real-time insights.

7.2.2 Automated Payment Gateway Integration

Integrate secure payment gateways like Stripe or PayPal for instant online donations and automatic receipt generation.

7.2.3 Mobile Application Version

Develop Android and iOS apps so donors and NGOs can interact with the system conveniently on mobile devices.

7.2.4 Two-Factor Authentication (2FA)

Strengthen security by adding OTP-based or email-based two-factor authentication during login.

7.2.5 AI-Based Donation Recommendation System

Implement AI to recommend campaigns to donors based on donation history, interests, and trending causes.

7.2.6 Multi-Language Support

Add languages such as Urdu to make the platform more accessible to local NGOs and donors.

7.2.7 NGO Progress Analytics

Provide NGOs with detailed visualization tools to track campaign progress and donor responses.

By implementing these enhancements, Hope Givers can grow into a more powerful, secure, and intelligent donation platform that continues to support users and contributes significantly to charitable activities.

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